C291 - Spring 26 - System Programming with C & Unix
Assignment 10

Question 1: Student Records (40 Marks)

Write a program in C to store, retrieve and modify students' records in a text file. Your program will use two files, a records file (that keeps the records) and a command file (that contains commands to insert, delete or modify records and the associated data). A record consists of a student ID, name, and course code.

The command file contains a single digit(representing a command) on a line, followed by zero or more lines of data that may contain

- a new record to insert, or
- an ID to search or delete a record
- an ID and course to modify an existing record

The commands are as follows:

1: Insert all the records listed on the next lines in the records file

2: Search for the student with the ID on the next line from the records file and display the record, if present.

3: Display all records from the records file

4: Remove all the records of the student with the ID on the next line from the file

5: Modify the course code of the student with the ID on the next line in the file

Note: Your program will open the commands file for *reading* and the records file for *writing*/*editing*. In addition to modifying the records file based on the commands, display an appropriate confirmation message on the screen indicating the records inserted, deleted, or modified. *Insert the records in the records file in the ascending order of the student ID.* **While the records file is provided in text format, you may store it as a binary file if it helps you in editing.**

Commands file: input_p1_ex1.txt and input_p1_ex2.txt

Commands: Insert (1), Search (2), Display (3), Remove (4), Modify (5)

Records file: studentrecords1.csv

Note: There is no need to display the content of the overall file in the output it's mentioned in output just for your reference.

Example 1:

Consider the following command and records files

Commands file:

```
1
2115 John c212
2115 John c215
2000 Kayle c212
2000 Kayle c215
2
2115
3
4
2000
5
2012 c212 c291
```

Records file:

```
1221    Sam   c212
1221    Sam   c215
1287    Eric  c212
1287    Eric  c215
1865    Mary  c212
1865    Mary  c215
2012    Raj   c212
```

Output:

The following records were inserted into the records file.

```
2115 John c212
2115 John c215
2000 Kayle c212
2000 Kayle c215
```

The content of the records file after insertion:

```
1221    Sam   c212
1221    Sam   c215
1287    Eric  c212
1287    Eric  c215
1865    Mary  c212
1865    Mary  c215
```

2000 Kayle c212
2000 Kayle c215
2012 Raj c212
2115 John c212
2115 John c215

Searching for a student with student id: 2115

The following records were found for a student with student id: 2115
2115 John c212
2115 John c215

The contents of the records file:
1221 Sam c212
1221 Sam c215
1287 Eric c212
1287 Eric c215
1865 Mary c212
1865 Mary c215
2000 Kayle c212
2000 Kayle c215
2012 Raj c212
2115 John c212
2115 John c215

Removing records of students with student id 2000
The contents of the records file after removal :
1221 Sam c212
1221 Sam c215
1287 Eric c212
1287 Eric c215
1865 Mary c212
1865 Mary c215
2012 Raj c212
2115 John c212
2115 John c215

The contents of the records file after modification:
1221 Sam c212
1221 Sam c215
1287 Eric c212
1287 Eric c215
1865 Mary c212
1865 Mary c215
2000 Kayle c212

2000 Kayle c215
2012 Raj c291
2115 John c212
2115 John c215

Example 2:

Input:

1
2101 Max c212
2000 Rachel c215
3
2
2101

Output:

Original records in the studentRecord csv file:
1221 Sam c212
1221 Sam c215
1287 Eric c212
1287 Eric c215
1865 Mary c212
1865 Mary c215
2012 Raj c212

The following records were inserted into the records file:
2101 Max c212
2000 Rachel c215

The content of the records file after insertion:
1221 Sam c212
1221 Sam c215
1287 Eric c212
1287 Eric c215
1865 Mary c212
1865 Mary c215
2000 Rachel c215
2012 Raj c212
2101 Max c212

The contents of the records file:

```
1221    Sam    c212
1221    Sam    c215
1287    Eric   c212
1287    Eric   c215
1865    Mary   c212
1865    Mary   c215
2000    Rachel c215
2012    Raj    c212
2101    Max    c212
```

Searching for a student with student id: 2101

The following records were found for a student with student id: 2101

```
2101    Max    c212
```

Problem 2: File Manipulation (20 Marks)

Write a C program where you have to perform the below operations on the two text files i.e **input21.txt** and **input22.txt** provided.

1. Remove empty lines from both files.
2. Merge the contents of 1st file with 2nd file.

Example 1:

Your input file name should be “input21.txt” and “input22.txt”

Contents of the input21.txt file are:

“
This is a good day.

How are you?
”

Contents of the input22.txt file are:

“
Go Hoosiers.

Hoosiers are the best.
”

Output:
After removing empty lines from both files:

Contents of **input21.txt**

“
This is a good day.
How are you?
“

Contents of **input22.txt**

“
Go Hoosiers.
Hoosiers are the best.
“

After merging **input21.txt** with **input22.txt**. Your **input22.txt** should look like

“
Go Hoosiers.
Hoosiers are the best.
This is a good day.
How are you?
“

Problem 3: AND, OR, and XOR pairs (20 Marks)

Write a C program to input an array arr[] of size N. The task is to find the number of pairs (arr[i], arr[j]) as AND_Pairs, OR_pairs and XOR_pairs such that:

- AND_pairs: Count of pairs where bitwise AND of least significant bits is 1.
- OR_pairs: Count of pairs where bitwise OR of least significant bits is 1.
- XOR_pairs: Count of pairs where bitwise XOR of least significant bits is 1.

You also need to print the corresponding pairs while printing the count.

Example1:

Input:

arr[] = {1, 2, 3}

Output:

Total AND pairs: 1
(1, 3)
Total OR pairs: 3
(1, 2)
(2, 3)
(1, 3)
Total XOR pairs: 2
(1, 2)
(2, 3)

Explanation:

Array in binary representation: arr[] = {01,10,11}

- AND of LSB of 1 and 2 : (01 & 10) = 0
- AND of LSB of 1 and 3 : (01 & 11) = 1
- AND of LSB of 2 and 3 : (10 & 11) = 0

- OR of LSB of 1 and 2 : (01 | 10) = 1
- OR of LSB of 1 and 3 : (01 | 11) = 1
- OR of LSB of 2 and 3 : (10 | 11) = 1

- XOR of LSB of 1 and 2 : (01 ^ 10) = 1
- XOR of LSB of 1 and 3 : (01 ^ 11) = 0
- XOR of LSB of 2 and 3 : (10 ^ 11) = 1

Your code should work for any array of any size. The above is just an example and not the only input.

Problem 4: Debugging (20 Marks)

Solve the following questions (both part a and b):

a) Given program is supposed to find the sum of the first N natural numbers. At the first sight the code looks fine, however the program provides incorrect results. Using GDB debugger find the source of the problem and submit a working program (after debugging). Also submit a screenshot(s) of GDB commands you used to figure out the problem while in the GDB terminal (you can put the screenshot(s) and explanation of how the commands you ran in GDB helped you figure out the bugs in a pdf file and submit along with the correct code). You can find the below code in solution_4_a.c file attached with the assignment.

```
#include <stdio.h>

int sum(int n) {
    int i;
    int result;
    for (i = 1; i <= n; i++) {
        result += i;
    }
    return result;
}

int main() {
    int n;
    printf("Enter n value: ");
    scanf("%d", & n);
    printf("Sum of n natural numbers is: %d\n", sum(n));
    return 0;
}
```

Submission files:

Correct code: solution_4_a.c
Pdf file (Screenshots, explanation, etc…): solution_4_a.pdf

b) Given program prompts for the user input, reads the input data and prints it on screen. However, due to the code containing one serious bug the executable crashes and reports segmentation fault. The output does not provide enough information to identify the source of the problem. Using GDB debugger find the source of the problem and submit a working program (after debugging). Also submit a screenshot(s) of GDB commands you used to figure out the problem while in the GDB terminal (you can put the screenshot(s) and explanation of how the commands you ran in GDB helped you figure out the bugs in a pdf file and submit along with the correct code). You can find the below code in solution_4_b.c file attached with the assignment.

```
#include <stdio.h>

int main() {
   int a = 0;
   printf("Please, input variable a\n");
   scanf("%d", a);
   printf("a: %d\n", a);
   return 0;
}
```

Submission files:

Correct code: solution_4_b.c
Pdf file (Screenshots, explanation, etc…): solution_4_b.pdf